Seokwon Jung

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Research Interests

I work on **agentic memory systems and information retrieval** for personalization: RAG systems, memory for reasoning, knowledge updating, and retrieval evaluation beyond semantic similarity. My recent work is a benchmark showing that current memory systems fail to keep knowledge current, resolvable only at impractical cost.

Education

KAIST, M.S. in Artificial Intelligence — Mar 2026 – Feb 2028 (expected)
Kim Jaechul Graduate School of AI; advised by Prof. Seong Joon Oh

KAIST, B.S. in Computer Science and Chemistry (double major) — Mar 2014 – Feb 2026

Publications

[1] **MEME: Multi-Entity & Evolving Memory Evaluation.**
Seokwon Jung, Alexander Rubinstein, Arnas Uselis, Sangdoo Yun, Seong Joon Oh.
NeurIPS 2026, Evaluations & Datasets Track. [pdf] [code]

[2] **Cross-Pollination Retrieval: Finding Transferable Ideas Across Fields in Scientific Literature.**
Seokwon Jung, Yerang Kim, Yejin Kim, Alexander Rubinstein, Arnas Uselis, Sangwoo Park, Seong Joon Oh, Sangdoo Yun.
Preprint, 2026. [pdf]

[3] **TULIP: Targeted LLM Unlearning at Layers Identified Per-Input.**
Yejin Kim, William F. Shen, **Seokwon Jung**, Daeun Park, Seong Joon Oh.
Preprint, 2026. [pdf]

[4] **Form Over Content in Gradient-Based Data Attribution Methods.**
Sunwoo Kim, **Seokwon Jung**, SoHyung Kim, Seong Joon Oh, Alice Oh.
Preprint, 2026. [pdf]

[5] **Break the Output Geometry for Large Language Model Unlearning.**
Yejin Kim, William F. Shen, **Seokwon Jung**, Seong Joon Oh.
ICML 2026 Workshop on the Impact of Memorization on Trustworthy Foundation Models. [pdf]

[6] **Polyphonic Sound Event Detection Using Convolutional Bidirectional LSTM and Synthetic Data-based Transfer Learning.**
Seokwon Jung, Jungbae Park, Sang Wan Lee.
ICASSP 2019, pp. 885–889. (41 citations) [pdf]

Research Experience

Bluesignum — Research Scientist — Jul 2024 – Aug 2024

- Set the **research direction and system design** for emotional speech synthesis and for coupling a conversational model to a speech pipeline

Humelo Inc. — Co-founder, Research Scientist — Apr 2018 – Apr 2024

- Research on **controllable speech synthesis**, taken from model to deployment

- Reached MOS 3.5+ by conditioning generation on emotion embeddings with an Arousal–Valence–Dominance formulation
- Extended a Neural Source Filter vocoder to multiple speakers via speaker embeddings and added pitch conditioning from a neural pitch tracker

- Research on **synthetic-data transfer learning** for polyphonic sound event detection [3]
  - Improved F1 by 28.4% over the prior architecture on TUT 2016
- Research on the **training corpus** that made expressive synthesis learnable
  - Defined 9 emotion categories over 40+ hours of speech, with an annotation protocol for emotional state and noise
  - Reused my sound event detection model as an automatic quality filter to strip noisy segments from the training set

Uneedcomms — Research Intern Oct 2017 – Feb 2018

- Built the **data collection pipeline** that made ML feasible, as no training data had previously been collected
  - Designed the schema and query layer from scratch, then automated performance reporting over it (Presto, Hadoop, Spark, Flask)

Patent

"Method for Training a Sound Event Detection Model" 2019

KR 10-2019-0007575; PCT/KR2019/012932.

Academic Service

Co-organizer — **PALM: Personalized, Aligned, Long-term Memory for AI Systems**, NeurIPS 2026 Workshop.

Skills

- **Languages:** Python, SQL, C++, Java
- **ML:** PyTorch, Hugging Face, DSPy, scikit-learn; LLM fine-tuning and evaluation, transfer learning, model interpretability
- **Retrieval:** sparse and dense retrieval (BM25, embeddings), cross-encoder reranking, RAG, vector stores (Qdrant), knowledge graphs (Neo4j)
- **Data:** Spark, Hadoop, Presto, Docker; synthetic data generation, LLM-as-judge pipelines with human calibration, A/B-style controlled ablations